Politecnico di Milano
School of Industrial and Information Engineering
Bayesian Learning and Monte Carlo Simulation

Final project:
Yacht Hydrodynamic Resistance

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1 Data and Problem

We analyse the *Yacht Hydrodynamics* dataset from the UCI Machine Learning Repository [1, 2]. The data come from towing-tank experiments on the Delft systematic yacht hull series. Each observation corresponds to one hull form tested at one speed. The response variable is the residuary resistance per unit weight of displacement, while the predictors describe two different aspects of the experiment: the speed, measured by the dimensionless Froude number $Fn = v/\sqrt{gL} \in [0.125, 0.45]$, and the hull geometry, summarized by five form factors: `LCB`, `prismatic`, `L_disp`, `beam_draught` and `L_beam` (Table 7). The hull variables are constant within a hull, whereas the Froude number changes across the towing-tank runs.

The scientific question is motivated by hydrodynamic theory. For sailing yachts, residuary resistance is expected to follow an approximate power law in the Froude number,

$$\text{resistance} \propto Fn^{\gamma_F},$$

with exponent close to 4 in the bulk regime. The project asks whether this relationship is visible in the data, which hull-geometry variables still matter after accounting for speed, and how the inference changes once the clear heteroscedasticity in resistance is modelled directly.

We address these questions in a Bayesian framework:

1. log–log regression to estimate and *verify* $\hat{\beta}_F \approx 4$, with a prior-sensitivity check;
2. BAS/BMA over the 2^5 hull-factor submodels, to see which carry signal once Fn is included;
3. a heteroscedasticity-corrected nonlinear model on the retained factors, contrasted with the log–log fit;
4. posterior predictive curves, with and without the heteroscedasticity treatment, against the data.

2 Log–log Regression

2.1 Model and prior

Taking logarithms turns the physical power law resistance $\propto Fn^{\gamma_F}$ into a linear model in which the exponent appears directly as a regression slope. We therefore fit

$$\log(\text{resistance}) = \beta_0 + \beta_F \log(\text{Froude}) + \sum_{k=1}^{5} \beta_k \,\text{hull}_k + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, \sigma^2), \quad (1)$$

where hull_k denotes the five geometric form factors; the smallest observed resistance is $0.01 > 0$, so the logarithm is well defined for every record and no offset is required. As a baseline we adopt the non-informative **Jeffreys reference prior**

$$\pi(\boldsymbol{\beta} \mid \sigma^2) \;\propto\; 1, \qquad \pi(\sigma^2) \;\propto\; \sigma^{-2}, \quad (2)$$

which lets the data dominate the inference; the sensitivity of the conclusions to an informative prior is examined in section 2.4.

2.2 Posterior and credible intervals

With the Jeffreys reference prior in Eq. (2), the Gaussian linear model has a closed-form posterior. Let $\hat{\boldsymbol{\beta}} = (\boldsymbol{X}^\top \boldsymbol{X})^{-1} \boldsymbol{X}^\top \boldsymbol{y}$. Then

$$\boldsymbol{\beta} \mid \sigma^2, \boldsymbol{y} \sim \mathcal{N}\Big(\hat{\boldsymbol{\beta}},\, \sigma^2\,(\boldsymbol{X}^\top \boldsymbol{X})^{-1}\Big)\,, \tag{3}$$

Integrating out σ^2 gives a multivariate Student-t posterior for $\boldsymbol{\beta}$ with $n-p$ degrees of freedom, so the 95% credible intervals coincide numerically with the classical OLS confidence intervals.

2.3 Results

Table 8 (in the appendix) reports the posterior summaries. The model explains the data very well ($R^2 = 0.97$, residual standard deviation $s = 0.32$ on the log scale) and the posterior of the Froude exponent is sharply concentrated,

$$\beta_F \approx 4.69, \qquad 95\%\ \mathrm{CI} = [4.60,\ 4.78],$$

confirming the expected power-law behaviour resistance $\sim Fn^4$ and verifying the project requirement $\hat{\beta}_F \approx 4$.

Fig. 5 (appendix) shows the log–log scatter with the posterior-mean line and its very tight 95% credible band (left), and the posterior of β_F (right): the whole posterior mass sits above the nominal value 4 ($\Pr(\beta_F > 4 \mid \boldsymbol{y}) \approx 1$), so the empirical exponent is best described as *slightly steeper* than the textbook value of four.

2.4 Sensitivity check: independent informative priors

The reference-prior analysis *verifies* $\beta_F \approx 4$ a posteriori; a complementary check *encodes* the physics in the prior and asks how strongly the prior must insist on 4 before it overrides the data. We replace Eq. (2) with **independent proper priors**,

$$\beta_F \sim \mathcal{N}(4, \tau^2), \qquad \beta_j \sim \mathcal{N}(0, 100^2)\ \ (j \neq F), \qquad \sigma^2 \sim \text{Inv-Gamma}(10^{-3}, 10^{-3}), \tag{4}$$

all components mutually independent. Since $\boldsymbol{\beta}$ and σ^2 are now independent a priori, unlike in the conjugate Normal–Inverse-Gamma family, the joint posterior is no longer available in closed form, but both full conditionals remain standard,

$$\boldsymbol{\beta} \mid \sigma^2, \boldsymbol{y} \sim \mathcal{N}\big(\boldsymbol{V}(\boldsymbol{S}_0^{-1}\boldsymbol{m}_0 + \boldsymbol{X}^\top \boldsymbol{y}/\sigma^2),\ \boldsymbol{V}\big), \qquad \boldsymbol{V} = (\boldsymbol{S}_0^{-1} + \boldsymbol{X}^\top \boldsymbol{X}/\sigma^2)^{-1}, \tag{5}$$

$$\sigma^2 \mid \boldsymbol{\beta}, \boldsymbol{y} \sim \text{Inv-Gamma}\big(a_0 + \tfrac{n}{2},\ b_0 + \tfrac{1}{2}\|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}\|^2\big), \tag{6}$$

where $\boldsymbol{m}_0$ and $\boldsymbol{S}_0$ collect the prior means and variances of Eq. (4). We therefore sample the posterior with a **Gibbs sampler** (10 000 iterations, 2 000 burn-in; the effective sample size for β_F exceeds 4 000 in every run) and sweep the prior strength $\tau \in \{0.05, 0.1, 0.2, 0.5\}$.

Table 9 (appendix) summarises the sweep; the posterior densities are overlaid in Fig. 8 (appendix). A moderately informative prior ($\tau = 0.5$) reproduces the reference result. Tightening the prior pulls the posterior mean towards 4, but only the near-dogmatic $\tau = 0.05$ (prior 95% interval $\approx [3.9, 4.1]$) drags it as far as 4.33, and even then $\Pr(\beta_F > 4 \mid \boldsymbol{y}) = 1$.

For any moderately diffuse physical prior ($\tau \geq 0.1$) the likelihood dominates and the posterior mean stays in 4.56–4.68. The conclusion that the empirical exponent is genuinely steeper than Fn^4 is therefore **robust to the prior**: it is a feature of the $n = 308$ observations, not an artefact of the non-informative choice, a transparent instance of prior–data conflict, in which an informative likelihood overwhelms all but an unreasonably confident prior.

3 BAS/BMA model selection over hull covariates

3.1 Model space and forced Froude effect

The log–log regression in section 2 establishes that the dominant physical scaling is the Froude power law. The model-selection question is therefore conditional on Froude: after log(Froude) has been included, do any hull-geometry variables still carry independent information about log(resistance)? For this reason the term log(Froude) is forced into every candidate model; the BAS/BMA step is not intended to test whether speed matters, but to identify possible secondary hull effects after speed has been accounted for.

The optional covariates are the five hull form factors `LCB`, `prismatic`, `L_disp`, `beam_draught` and `L_beam`. They are treated as optional because the full log–log regression does not resolve any of them individually: their credible intervals cross zero, and the form factors are strongly related across the limited set of 22 hull designs. With five optional variables, the model space contains only $2^5 = 32$ submodels. We can therefore enumerate the whole space with `bas.lm`, using `include.always = ~ log_Fr` and a uniform prior over the 32 hull-covariate submodels.

This averaging approach is preferable to interpreting only the full regression coefficients. The full model conditions on all five hull covariates being present simultaneously, even though they are correlated and weakly identified. BAS/BMA instead averages over all possible subsets and reports, for each covariate, its posterior inclusion probability (PIP): the posterior probability that the covariate belongs in the regression model, conditional on log(Froude) being included.

3.2 Baseline BAS/BMA results

The baseline analysis uses the BIC marginal-likelihood approximation in `bas.lm` with a uniform model prior. The resulting PIPs are shown in Table 1. A large PIP would indicate that a covariate appears repeatedly in high-posterior-probability models, while a small PIP means that the data do not require that term once Froude and the competing hull variables are considered.

Table 1: Approximate posterior inclusion probabilities for the five optional hull-geometry covariates when log(Froude) is forced into every model. Values are based on enumeration of the $2^5 = 32$ submodels using the BIC marginal-likelihood approximation, matching the baseline `prior = "BIC"` specification in `bas.lm`.

Covariate	PIP	Interpretation
`beam_draught`	0.57	moderate, not decisive
`prismatic`	0.27	weak
`LCB`	0.21	weak
`L_disp`	0.21	weak
`L_beam`	0.19	weak

The largest PIP is for `beam_draught`, approximately 0.57. This is moderate evidence, not decisive evidence: the variable is the most plausible secondary hull covariate under the baseline prior, but it is far from being overwhelmingly selected. The other four hull variables have PIPs well below 0.50, so their evidence is weak conditional on log(Froude). This agrees with the full-model coefficient intervals in Table 8, but is more defensible because it checks the whole model space rather than relying on a single saturated regression.

3.3 Sensitivity to prior choices

Because the baseline PIPs are modest, the BAS/BMA conclusion was checked under alternative BAS/BMA prior specifications. The same model space was enumerated in each run: log(Froude) was forced into every model, the five hull covariates were optional, and the model prior over the $2^5 = 32$ submodels was kept uniform. The prior specifications compared were the baseline `BIC / reference` approximation, Zellner's `g-prior (g = n)`, and the heavy-tailed `JZS` prior. The resulting PIPs are reported in Table 2.

Prior specification	LCB	prismatic	L_disp	beam_draught	L_beam
`BIC / reference`	0.210	0.265	0.206	0.565	0.192
`g-prior (g = n)`	0.186	0.217	0.182	0.499	0.166
`JZS`	0.060	0.057	0.051	0.243	0.042

Table 2: Sensitivity of posterior inclusion probabilities to alternative BAS/BMA prior specifications.

The sensitivity analysis reinforces the need for caution. The robust conclusion is not that `beam_draught` is definitely selected. In fact, no hull covariate reaches PIP ≥ 0.50 under all three prior specifications, and the magnitude of the inclusion probabilities changes noticeably. The robust conclusion is instead that all hull covariates are much weaker than the forced Froude effect once log(Froude) is included. At the same time, `beam_draught` remains the best-supported secondary covariate in the baseline analysis and the top-ranked hull covariate in each sensitivity run, so it remains a reasonable candidate rather than an arbitrary choice. The observed prior sensitivity motivates a cautious interpretation; it does not invalidate the analysis, but it prevents any strong claim that a hull covariate has been unambiguously selected.

3.4 Interpretation and link with the nonlinear model

Overall, BAS/BMA suggests that `beam_draught` is the most defensible secondary covariate, while the Froude power law remains the dominant robust signal. This gives a coherent reduced modelling strategy for the next section: the heteroscedastic nonlinear analysis should compare a Froude-only model M_0 with a model M_1 that also includes `beam_draught`. In this workflow, BAS/BMA proposes the candidate hull covariate, and the nonlinear original-scale model checks whether that covariate improves posterior predictive performance once the variance growth with resistance is modelled explicitly.

4 Heteroscedasticity-Corrected Nonlinear Model

4.1 Nonlinear model

Following the BAS/BMA analysis, the only hull-form variable carried forward was the beam-draught ratio, whose posterior inclusion probability was about 0.565, so the evidence was moderate rather than strong. Therefore, we fitted both a Froude-only model **M0** and a model including beam-draught **M1**. Because the residuary resistance is strictly positive, the main heteroscedastic nonlinear model uses a **Gamma** likelihood, parametrized through its mean and variance,

$$Y_i \mid \theta, \sigma_0, \delta \sim \text{Gamma}\Big(\text{shape}_i = \tfrac{\mu_i^2}{\sigma_i^2},\ \text{rate}_i = \tfrac{\mu_i}{\sigma_i^2}\Big), \qquad \mathbb{E}(Y_i) = \mu_i, \quad \text{Var}(Y_i) = \sigma_i^2,$$

where Y_i is the observed residuary resistance and

$$\mu_i = k\,\text{Froude}_i^{\gamma_F} \exp\{\beta_B\,\text{bd}_i\}.$$

Here bd_i is the beam–draught ratio (`beam_draught`), and $k > 0$ is the multiplicative constant of the power law. To model heteroscedasticity, we did not assume a constant variance. Instead, we used

$$\sigma_i = \sigma_0 \mu_i^{\delta}.$$

The parameter δ controls how fast the standard deviation grows with the fitted mean. If $\delta = 0$, then the variance is constant. If $\delta = 1$, then the standard deviation is proportional to the mean (constant coefficient of variation), which is close to the multiplicative-error interpretation of the log-log model.

The parameters to be estimated are the constant k, the Froude exponent γ_F, the hull coefficient β_B, the baseline noise level σ_0, and the **variance-growth exponent** δ. We also test the heteroscedasticity correction by comparing the fixed-variance version ($\delta = 0$) with the learned-variance version using WAIC and PSIS-LOO.

For computation, the mean was rewritten in exponential form,

$$\mu_i = \exp\{\log k + \gamma_F \log(\text{Froude}_i) + \beta_B \text{bd}_i\},$$

so that it is a log-linear predictor in the parameters, and the Gamma shape and rate are obtained from μ_i and σ_i by moment matching as above.

4.2 Prior distributions

All priors are weakly informative and mutually independent. For the mean parameters,

$$\log k \sim \mathcal{N}(0, 10^2), \qquad \gamma_F \sim \mathcal{N}(4, 2^2), \qquad \beta_B \sim \mathcal{N}(0, 2^2). \tag{7}$$

The prior on γ_F is centred at the physical value 4. For the variance function,

$$\log \sigma_0 \sim \mathcal{N}(0, 10^2), \qquad \delta \sim \mathcal{N}(0, 1^2). \tag{8}$$

The prior on δ is centred at 0, i.e. at homoscedasticity, precisely so that any evidence of variance growth comes from the data and not from the prior. As a sensitivity check on the headline quantity, the γ_F prior is also replaced by a vague $\mathcal{N}(0, 10^2)$ and by an informative $\mathcal{N}(4, 0.5^2)$.

4.3 Posterior sampling algorithm

Unlike the log-log linear model, the posterior distribution is not available in closed form. The likelihood contains a nonlinear exponential mean, and the variance also depends on the mean through $\sigma_i = \sigma_0 \mu_i^\delta$. Because of this non-conjugacy, we used MCMC in JAGS.

For this model JAGS automatically assigned the univariate slice sampler to the continuous parameters.

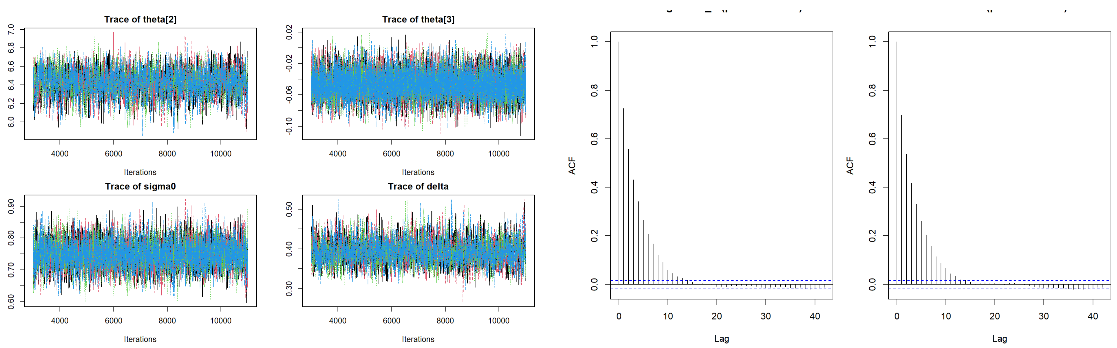
We run four independent chains, started from dispersed values of γ_F $(3, 6, 5, 4)$ and of δ $(0, 1, 0.5, 1.2)$. Each chain uses an adaptation phase, a burn-in that is discarded, and a thinned sampling phase. If chains launched from very different points all settle on the same distribution, that is good evidence the sampler has reached the posterior rather than a starting-value-dependent region.

4.4 Convergence diagnostics

Before doing any interpretation from the posterior distribution or sampling from the posterior, we should check whether our proposed model has converged. We checked the convergence of the nonlinear model with several different methods. First we check the Gelman–Rubin statistic $\hat{R}$. This compares the variance between the dispersed chains to the variance within each chain. If the chains have all converged to the same distribution the two variances agree. Secondly, we check Effective sample size (ESS). Successive MCMC draws are autocorrelated, so N kept draws carry less information than N independent ones. The ESS estimates the equivalent number of independent draws; we want it in the thousands so that posterior means and quantiles are estimated with little Monte-Carlo error. Trace plots, for each parameter the four chains should look like stationary noise fluctuating around a common, fixed level, with the chains overlapping rather than drifting or sitting at separate values. Finally, Autocorrelation function (ACF), For each parameter the autocorrelation of the chain should decay to zero within a few lags.

Table 3: Convergence diagnostics for the heteroscedastic nonlinear model M1 under the main prior.

Parameter	$\hat{R}$	Effective sample size
δ	1.0004	3594
σ_0	1.0000	6710
$\log k$	1.0007	3328
γ_F	1.0008	3106
β_B	1.0006	12000



(a) Trace plots **(b)** Autocorrelation functions

Figure 1: Trace plots and autocorrelation functions for model M1; each colour represents a chain.

Table 3 reports the convergence diagnostics for the main M1 run. All Gelman–Rubin statistics are essentially equal to one, indicating that the independently initialized chains are sampling from the same posterior distribution. The effective sample sizes are also sufficiently large for posterior inference, especially for β_B. Figure 1 shows that the different chains overlap on the plot and look like stationary noise; it shows that every chain mixed with the given sampling numbers, and the ACF decays to zero within a few lags.

4.5 Posterior analysis

Table 4 reports the posterior summaries for M1, and the marginal posteriors of the three parameters of interest are shown in Fig. 2.

Table 4: Posterior summaries for the heteroscedastic nonlinear model M1 (Froude + `beam_draught`) with the Gamma likelihood, pooled over four chains; k is the multiplicative constant of the power law.

Parameter	**Mean**	**SD**	**2.5%**	**50%**	**97.5%**
k	1446.56	223.93	1053.74	1431.21	1927.33
γ_F	4.8168	0.0647	4.6910	4.8155	4.9469
β_B	0.0501	0.0314	−0.0104	0.0500	0.1123
σ_0	0.3527	0.0171	0.3213	0.3522	0.3880
δ	0.8881	0.0246	0.8388	0.8884	0.9356

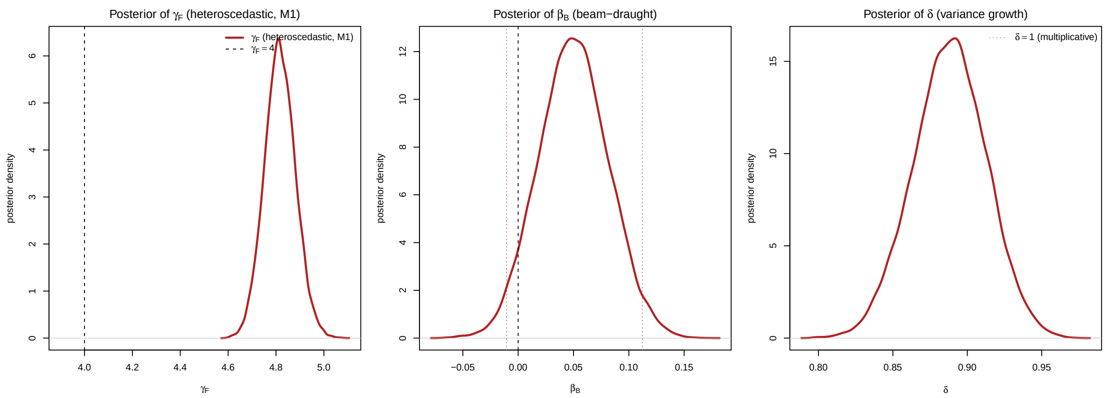


Figure 2: Marginal posteriors for the heteroscedastic nonlinear model M1 (Gamma likelihood). Left: the Froude exponent γ_F, with the dashed line at the physical value 4. Centre: the beam–draught coefficient β_B, with the dashed line at 0. Right: the variance-growth exponent δ, with the multiplicative value $\delta = 1$ marked.

The Froude exponent is estimated at $\gamma_F \approx 4.82$, with a 95% credible interval $[4.70, 4.95]$ that lies entirely above the physical value 4 ($\Pr(\gamma_F > 4 \mid \boldsymbol{y}) \approx 1$). This is very close to the log-log estimate $\beta_F \approx 4.69$ of section 2, so the original-scale Gamma model and the log-log model agree on the power-law exponent. The variance-growth exponent is $\delta \approx 0.89$, credibly positive ($\Pr(\delta > 0 \mid \boldsymbol{y}) \approx 1$) and close to 1, which means the standard deviation grows almost in proportion to the mean. The data are therefore clearly heteroscedastic, and the variance grows almost as fast as the multiplicative assumption that is implicit in the log-log model.

The hull covariate is weak under this model. The beam-draught coefficient is $\beta_B \approx 0.05$, but its 95% credible interval $[-0.013, 0.113]$ contains zero ($\Pr(\beta_B < 0 \mid \boldsymbol{y}) \approx 0.06$), so it is not credibly different from zero. In other words, once the positive-support Gamma likelihood is used, the moderate BAS evidence for `beam_draught` does not become a credible physical effect.

We also performed a prior sensitivity check for γ_F to test the proposed physical value 4. The main prior is $\gamma_F \sim \mathcal{N}(4, 2^2)$. To see whether the posterior estimate was mainly driven by this prior, we repeated the analysis with a vague prior, $\mathcal{N}(0, 10^2)$, and with a more informative prior, $\mathcal{N}(4, 0.5^2)$. The posterior mean was about 4.82 under the vague prior and about 4.80 under the informative prior, essentially unchanged from the main prior. This shows that the conclusion $\gamma_F \approx 4.8$ is driven by the likelihood and the data, not by the prior.

Table 5: Prior sensitivity of the Froude exponent γ_F in M1 under three priors. The posterior is essentially unchanged, so the result is likelihood-driven.

Prior on γ_F	**Mean**	**2.5%**	**97.5%**
$\mathcal{N}(4, 2^2)$ (main)	4.8191	4.6957	4.9509
$\mathcal{N}(0, 10^2)$ (vague)	4.8184	4.6953	4.9522
$\mathcal{N}(4, 0.5^2)$ (informative)	4.8021	4.6801	4.9332

4.6 Additional Sensitivity Checks

We have concluded three aditional tests on the model and likelihood selections. First, we have compared the Froude only mean structure M0 against M1, which add the `beam_draught`. Under the gamma likelihood the two models are essentially tied, WAIC for M1 model improves by only 0.3, and the LOO difference, $\Delta\text{elpd} = -0.2 \pm 1.9$. Furthermore, we have tested the same M1 mean with a constant variance ($\delta \equiv 0$) against the learned variance. The learned-variance model is overwhelmingly preferred: WAIC drops by more than 800 and $\Delta\text{elpd} = -413.7 \pm 14.2$, about thirty standard errors. Lastly, as a likelihood-sensitivity check we refit the identical M1 mean $\mu_i = \exp(\boldsymbol{x}_i^\top \boldsymbol{\theta})$ and the identical learned scale $\sigma_i = \sigma_0 \mu_i^\delta$ under an additive Normal likelihood, $Y_i \sim \mathcal{N}(\mu_i, \sigma_i^2)$, which is the most literal reading of the additive error in the project formula. The priors and the free δ are identical, so the likelihood family is the only thing that changes, and because both models are densities of the same observable on the same original scale their WAIC/PSIS-LOO are directly comparable.

Table 6: Likelihood sensitivity for M1: Gamma main model vs. additive Normal, same mean and learned-variance structure. WAIC/LOOIC are on the original resistance scale; the predictive columns are evaluated at the observed design points. The two fit the data equally well; they differ in support and in the inference on γ_F and `beam_draught`.

Likelihood	**WAIC**	**LOOIC**	$\max \hat{k}$	**95% cov.**	**band width**	$\Pr(y^{\text{rep}} < 0)$
Gamma	840.6	840.7	0.4	0.961	9.66	0.000
Additive Normal	842.8	842.8	0.6	0.987	5.55	0.127

$\Delta\text{elpd} = -1.1, \quad \text{se}_\Delta = 16.9$ (Normal vs. Gamma — not significant).

Table 6 reports the result. On fit the two are a tie: WAIC differs by only about 2, and $\Delta\text{elpd} = -1.1$ against $\text{se}_\Delta = 16.9$, well within Monte-Carlo noise; all Pareto-$\hat{k} < 0.6$, so this null result is reliable. The data therefore cannot prefer one likelihood over the other on fit alone. They differ in two other ways. First, the additive Normal places about 13% of its posterior-predictive mass below zero, which is impossible for a resistance, while the Gamma places none; the Normal does give sharper bands (mean 95% width 5.5 vs. 9.7), but partly through this negative tail. Second, and more important, the inference itself changes: the additive Normal inflates the Froude exponent to $\gamma_F \approx 6.4$ and forces `beam_draught` credibly negative, neither of which survives under the Gamma. We keep the Gamma as the main model, both because it respects the support and because its $\gamma_F \approx 4.8$ agrees with the independent log-log estimate, and we read the additive Normal as evidence that the steep exponent and the negative hull effect are artifacts of the error model rather than robust physical effects.

5 Posterior Predictive Curves

The final comparison is made on the original resistance scale, because this is where the data are observed and where the two error assumptions imply visibly different uncertainty bands. For a fair comparison, both curves in Fig. 3 are posterior predictive curves: each posterior draw includes uncertainty in the mean curve and one new observation-level error term. The observed yacht-resistance data are overlaid on the same scale.

For the log–log regression, the posterior predictive distribution is lognormal after transforming back to resistance. After drawing $(\boldsymbol{\beta}, \sigma^2)$ from the posterior, replicated values are generated as

$$\log y_i^{\text{rep}} \sim \mathcal{N}(x_i^\top \boldsymbol{\beta}, \sigma^2), \qquad y_i^{\text{rep}} = \exp(\log y_i^{\text{rep}}).$$

For the heteroscedastic nonlinear M1 model, replicated values are generated on the original scale with a mean-dependent standard deviation,

$$y_i^{\text{rep}} \sim \mathcal{N}(\mu_i, \sigma_i^2), \qquad \mu_i = \exp\{b_0 + \gamma_F \widetilde{\log Fn}_i + \beta_B \widetilde{bd}_i\}, \qquad \sigma_i = \sigma_0 \mu_i^\delta.$$

The displayed curves hold the hull form fixed at a representative value: `beam_draught` is set to its sample median, while the other log–log hull covariates are set to their sample means. This isolates the Froude–resistance relationship instead of mixing different hull geometries.

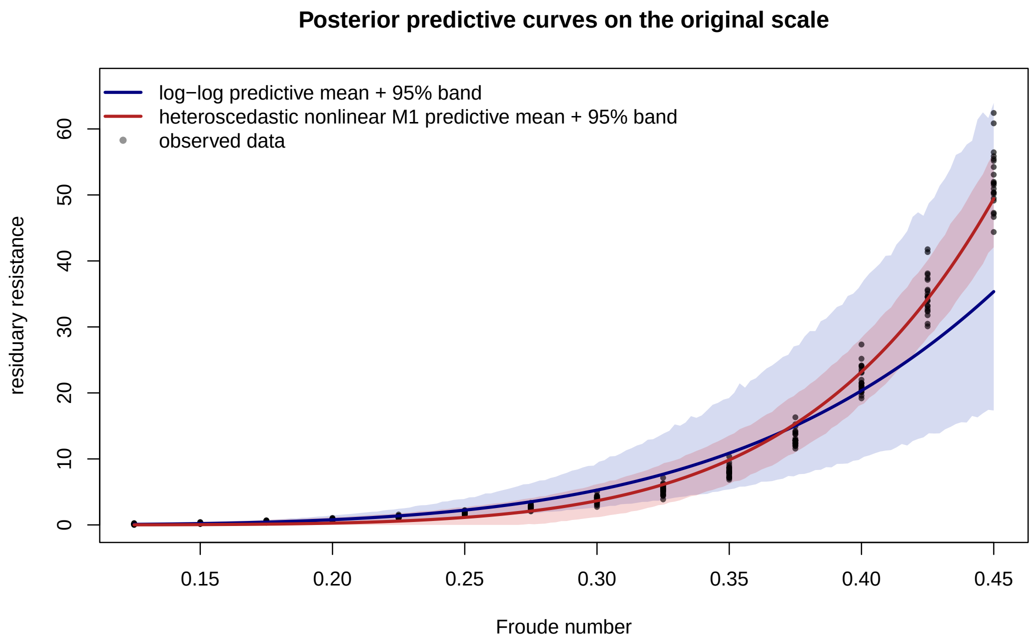


Figure 3: Posterior predictive curves on the original resistance scale. The blue band is the log–log model after back-transformation, and the red band is the heteroscedastic nonlinear M1 model. Points are the observed yacht resistance data.

The log–log fit keeps the Froude exponent close to the physical fourth-power law and produces multiplicative uncertainty. Its band is tight near zero resistance and expands as resistance grows, which describes the low and middle Froude range reasonably well. However, the mean curve is too flat at the highest Froude values, where the observed resistance rises faster than the fitted $Fn^{4.69}$ relationship.

The heteroscedastic nonlinear fit behaves differently. Its mean curve is steeper ($\gamma_F \approx 6.4$) and follows the high-Froude observations more closely. Because its variance is learned through $\sigma_i = \sigma_0 \mu_i^\delta$ with $\delta \approx 0.4$, the predictive band is not forced to have constant width. It widens where resistance and scatter are larger and tightens toward the low-Froude region. This is the practical payoff of the heteroscedasticity treatment: the uncertainty band adapts to the data rather than imposing a single residual scale across the whole Froude range.

Overall, the posterior predictive curves show that the two models differ mainly in how they scale uncertainty with resistance. The log–log model gives a useful positive predictive distribution, but it imposes multiplicative uncertainty and misses part of the high-speed curvature. The heteroscedastic nonlinear model learns the variance growth directly, tracks the observed data more closely, and supports the conclusion that the treatment of heteroscedasticity is essential for interpreting both the Froude exponent and the selected hull covariate `beam_draught`.

6 Train/Test Validation

As an additional predictive check, we also split the data randomly into training and test sets, using about 80% of the observations for training and 20% for testing. This validation is not used to replace the full-data Bayesian analysis: the scientific conclusions above are based on the posterior fitted to all 308 observations. The split is only a held-out check of whether the posterior predictive distributions behave sensibly on unseen rows.

The validation is kept on the same Bayesian scale as the posterior predictive curves. Instead of comparing only point predictions through RMSE or MAE, each candidate model is evaluated through its posterior predictive distribution on the test rows. For the log–log model, the predictive density is lognormal on the original resistance scale. For the nonlinear models, the predictive density is normal on the original scale, either with constant residual variance or with the learned heteroscedastic variance $\sigma_i = \sigma_0 \mu_i^{\delta}$.

For each test observation we compute the posterior predictive log density,

$$\log p(y_i^{\text{test}} \mid y^{\text{train}}),$$

and then summarize the models by total log predictive density, mean log predictive density, 95% predictive-interval coverage, and average predictive band width. Higher log predictive density indicates better held-out predictive accuracy. Coverage close to 95% indicates calibrated uncertainty, while the band width checks whether good coverage is achieved efficiently rather than by making intervals unnecessarily wide.

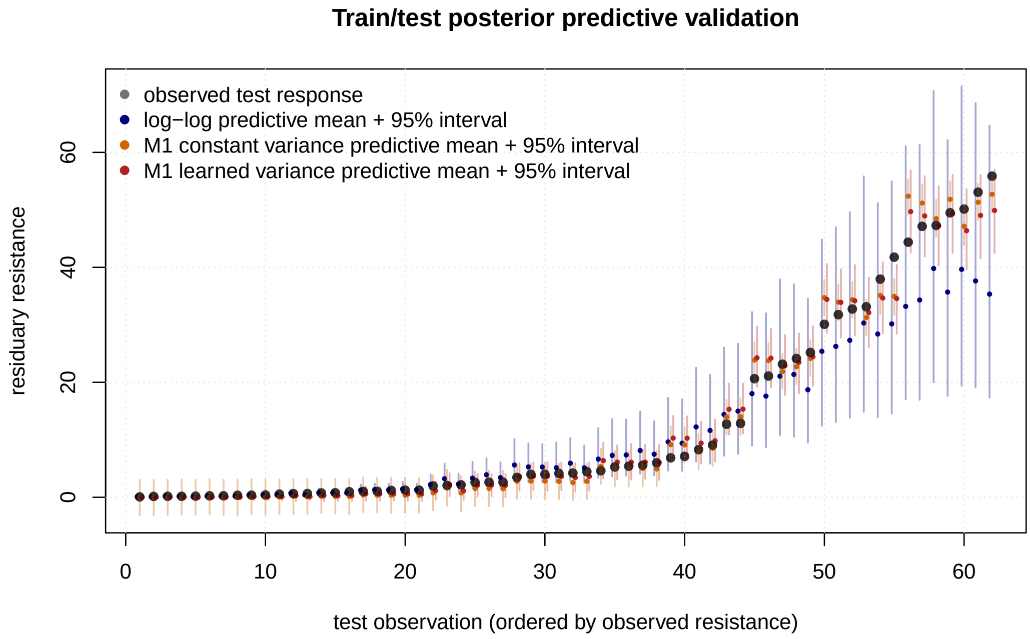


Figure 4: Train/test posterior predictive validation on the held-out rows. The black points are observed test responses ordered by resistance. Coloured points show posterior predictive means, and vertical intervals show the corresponding 95% posterior predictive bands for the log–log model, the constant-variance nonlinear M1 model, and the learned-variance nonlinear M1 model.

This validation supports the same interpretation as the posterior predictive curves. The preferred model is the nonlinear M1 specification with `beam_draught` and learned heteroscedastic variance when it combines strong held-out log predictive density with calibrated 95% intervals and reasonable band width. In other words, the validation checks both the location of the fitted curve and the uncertainty around it, confirming that the heteroscedasticity treatment improves prediction without reducing the comparison to a non-Bayesian point-fit exercise.

References

[1] I. Ortigosa, R. Lopez, and J. Garcia. *Yacht Hydrodynamics*. UCI Machine Learning Repository. DOI: 10.24432/C5XG7R. 2007.

[2] J. Gerritsma, R. Onnink, and A. Versluis. *Geometry, Resistance and Stability of the Delft Systematic Yacht Hull Series*. Technical Report. Delft University of Technology, Ship Hydromechanics Laboratory, 1981.

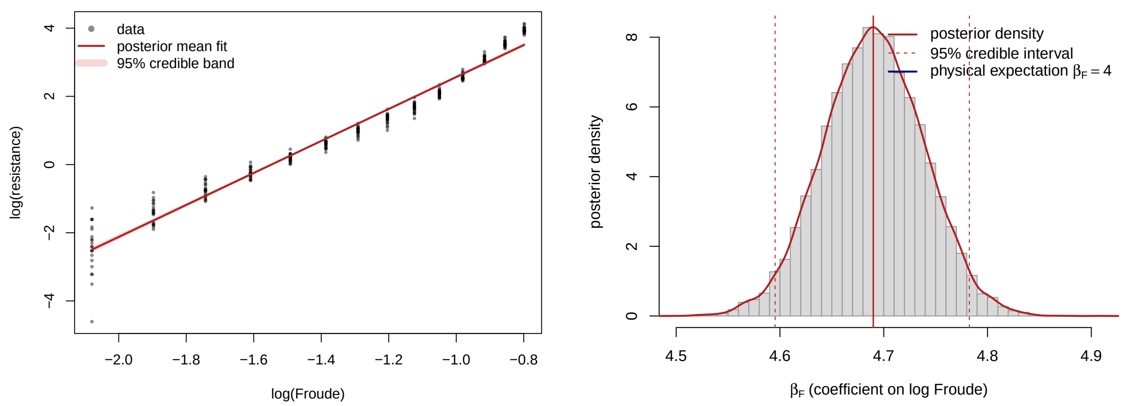
A Tables and Figures

Table 7: Variables of the yacht hydrodynamics dataset.

Variable	Description
`LCB`	Longitudinal position of the centre of buoyancy
`prismatic`	Prismatic coefficient
`L_disp`	Length–displacement ratio
`beam_draught`	Beam–draught ratio
`L_beam`	Length–beam ratio
`Froude`	Froude number, $Fn = v/\sqrt{gL}$
`resistance`	Residuary resistance per unit weight of displacement

Table 8: Posterior summary of the log–log model in Eq. (1) under the reference prior. Point estimates and 95% credible intervals coincide with the OLS fit (`lm`/`confint`); Pr(> 0) is the posterior probability that the coefficient is positive.

Coefficient	**Mean**	**SD**	**2.5%**	**97.5%**	Pr(> 0)
Intercept β_0	7.183	0.983	5.248	9.119	1.00
log(Froude) β_F	4.690	0.048	4.596	4.785	1.00
`LCB`	0.022	0.012	−0.003	0.046	0.96
`prismatic`	−0.621	1.601	−3.771	2.529	0.35
`L_disp`	0.465	0.513	−0.546	1.475	0.82
`beam_draught`	−0.066	0.200	−0.460	0.327	0.37
`L_beam`	−0.464	0.515	−1.477	0.549	0.18



(a) Power-law fit on log–log axes. **(b)** Posterior of β_F.

Figure 5: Left: observed log(resistance) against log(Froude) with the posterior-mean regression line (hull covariates held at their means) and its 95% credible band. Right: Student-t posterior density of the Froude exponent β_F; the dashed lines mark the 95% credible interval and the solid blue line the physical reference value $\beta_F = 4$.

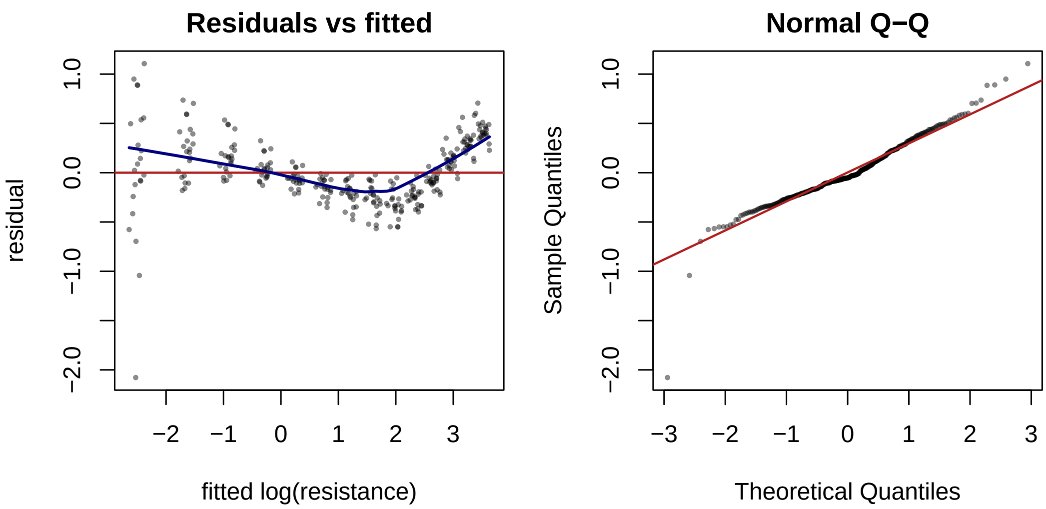


Figure 6: Residual diagnostics for the log–log model. Left: residuals versus fitted values, with a LOWESS trend (blue) revealing speed-dependent curvature and a variance that shrinks as the fitted value grows. Right: Normal Q–Q plot, showing heavier-than-Gaussian lower tails.

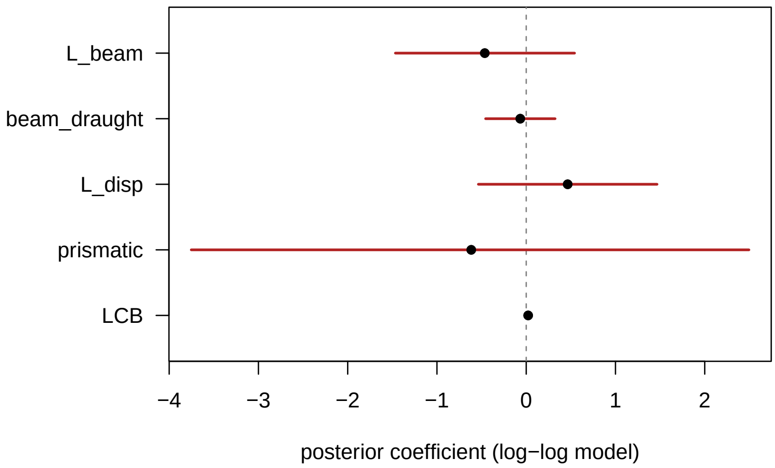


Figure 7: Posterior means and 95% credible intervals of the five hull-geometry coefficients in the log–log model. Every interval straddles zero (dashed line), indicating that no single form factor is resolved once the Froude number is accounted for.

Hull-geometry covariates. Once the Froude number is in the model, none of the five hull form factors is individually credible: every 95% interval comfortably contains zero (Table 8 and Fig. 7). The form factors vary only across the 22 hulls and are strongly collinear — the `prismatic` coefficient, for instance, has a posterior standard deviation of about 1.6. This near-redundancy is precisely what the BAS/BMA model selection in section 3 resolves, through posterior inclusion probabilities rather than single-model p-values.

Residual diagnostics. Although the global fit is excellent, the residuals (Fig. 6) reveal two systematic defects. First, a ∪-shaped trend against the fitted values: a single constant exponent slightly over-predicts at intermediate speeds and under-predicts at the highest speeds. Second, the residual spread is markedly larger at low fitted values, i.e. at low Froude numbers where the resistance is tiny and dominated by measurement noise — a textbook signature of heteroscedasticity, also visible as the fanning of the data cloud on the left of Fig. 5. Both findings directly motivate the heteroscedasticity-corrected nonlinear model developed in the main text.

Table 9: Posterior of the Froude exponent β_F under the independent informative priors in Eq. (4) of increasing strength, obtained by Gibbs sampling, compared with the closed-form reference-prior result.

Prior on β_F	**Mean**	**2.5%**	**97.5%**	$\Pr(\beta_F > 4)$
$\mathcal{N}(4, 0.05^2)$	4.33	4.25	4.41	1.00
$\mathcal{N}(4, 0.10^2)$	4.56	4.47	4.64	1.00
$\mathcal{N}(4, 0.20^2)$	4.65	4.56	4.75	1.00
$\mathcal{N}(4, 0.50^2)$	4.68	4.59	4.78	1.00
reference $\pi(\boldsymbol{\beta}, \sigma^2) \propto \sigma^{-2}$	4.69	4.60	4.78	1.00

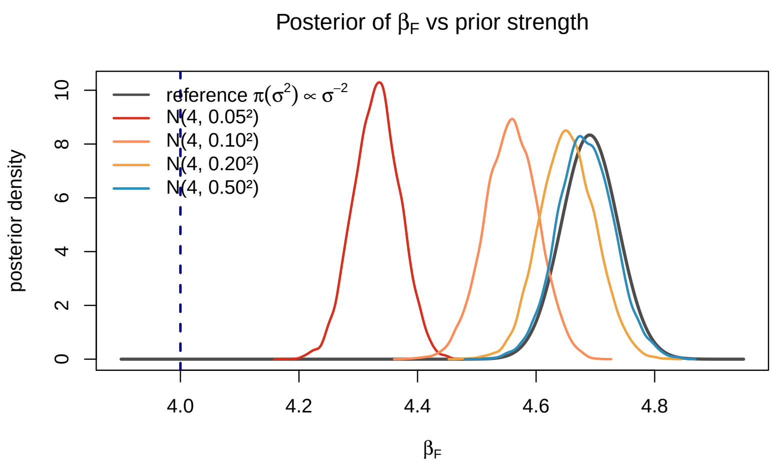


Figure 8: Posterior density of the Froude exponent β_F under the independent informative priors $\mathcal{N}(4, \tau^2)$ of increasing strength (coloured, Gibbs samples) and under the reference prior (grey, closed-form Student-t). The dashed blue line marks the physical value $\beta_F = 4$. Tightening the prior toward 4 shifts the posterior only modestly; the data keep it well above 4 unless the prior is unreasonably confident.

Table 10: Mean-structure check (Check 1): Froude-only M0 vs. M1 with `beam_draught`, both with learned variance. Δelpd and se_Δ are from `loo_compare` against M1.

Model	WAIC	LOOIC	p_{WAIC}	$\max \hat{k}$	ΔWAIC	Δelpd (se_Δ)
M0: Froude only	840.9	841.0	5.8	0.6	0.3	-0.2 (1.9)
M1: + `beam_draught`	840.6	840.7	7.1	0.4	0.0	0.0 (–)

Table 11: Variance-structure check (Check 2): same M1 mean with constant variance ($\delta = 0$) vs. learned variance (δ free). Δelpd and se_Δ are from `loo_compare` against the learned-variance model.

Variance	WAIC	LOOIC	p_{WAIC}	$\max \hat{k}$	ΔWAIC	Δelpd (se_Δ)
Constant ($\delta = 0$)	1668.0	1668.0	4.0	0.3	827.4	-413.7 (14.2)
Learned (δ free)	840.6	840.7	7.1	0.4	0.0	0.0 (–)

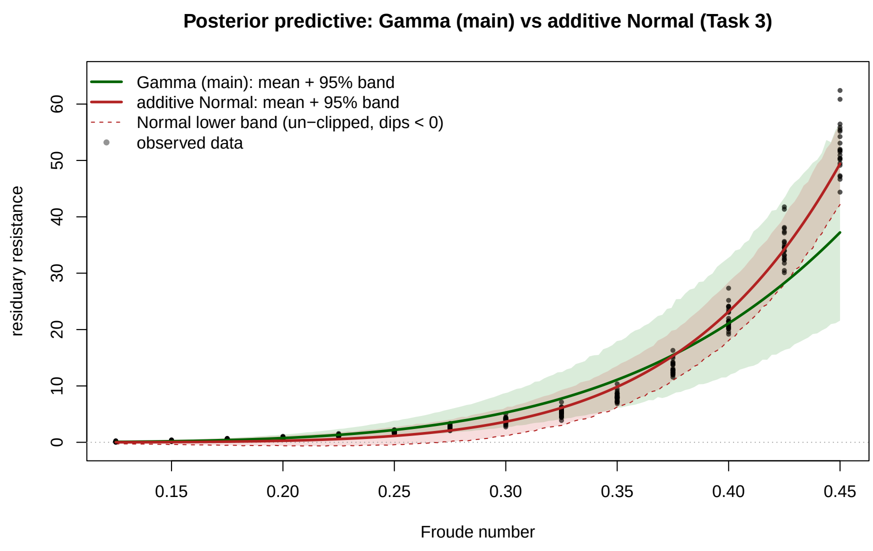


Figure 9: Posterior-predictive bands of the Gamma main model (green) and the additive-Normal sensitivity fit (red) on the original resistance scale, with observed data. The additive Normal's mean curve is steeper and its un-clipped lower band (dashed) dips below zero at low Froude, whereas the Gamma's lower band stays non-negative.